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Statistical Computing

BCS051

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** CSE, IT | **Semester:** 5 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 3rd year, Departmental Elective, 2024-25 onwards

UNIT 1 Descriptive Statistics and Probability

UNIT 2 Inferential Statistics and Linear Methods

UNIT 3 Random Numbers, Monte Carlo and MCMC

UNIT 4 Resampling, Density Estimation and Numerical Methods

UNIT 5 Statistical Computing with R

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: formulas, methods and comparison tables.
3. **Solved examples**: the numericals that appear (mean and SD, correlation and regression, distributions, confidence intervals, t, chi-square and ANOVA tests, Monte Carlo, Newton-Raphson, Simpson's rule).
4. **Formulas / code to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (central tendency, dispersion, skewness, correlation, regression, Bayes, distributions, Markov and Chebyshev, CLT) and Unit 2 (confidence intervals, hypothesis tests, ANOVA, chi-square, regression by QR and SVD, PCA).
- **Afternoon (3 hours):** Unit 3 (inverse transform, acceptance-rejection, Monte Carlo integration and variance reduction, Metropolis-Hastings and Gibbs) and Unit 4 (bootstrap, jackknife, cross-validation, kernel density, Newton-Raphson, numerical integration, EM, simplex).
- **Evening (2 hours):** Unit 5: R objects, functions, loops, import/export, plots and the built-in tests. Then the "Last-hour checklist".
- **In the exam:** state the hypotheses, the test statistic, its value, the critical value and the decision. For algorithms, write numbered steps and a short R snippet.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Descriptive Statistics and Probability

✍️ 2-mark definitions

- **Descriptive statistics:** methods for summarizing and presenting data (tables, charts, averages, spread).
- **Diagrammatic representation:** bar charts, pie charts, histograms, frequency polygons, ogives, box plots and stem-and-leaf plots.
- **Measures of central tendency:** mean (sum/count), median (middle value when sorted), mode (most frequent value).
- **Measures of dispersion:** range, quartile deviation, mean deviation, variance, standard deviation, coefficient of variation (CV = σ/μ × 100).
- **Skewness:** lack of symmetry. Positive skew: long right tail, mean > median > mode. Karl Pearson's coefficient Sk = (mean − mode)/σ ≈ 3(mean − median)/σ.
- **Kurtosis:** peakedness of a distribution compared with the normal: β2 = μ4/μ2²; β2 = 3 mesokurtic, > 3 leptokurtic, < 3 platykurtic.
- **Correlation coefficient (r):** a measure of linear association between −1 and +1.
- **Regression:** fitting an equation to predict one variable from others; simple linear regression y = a + bx.
- **Multiple correlation:** correlation between one variable and the best linear combination of several others.
- **Probability:** a measure P on events with 0 ≤ P(A) ≤ 1, P(S) = 1 and additivity for disjoint events (Kolmogorov axioms).
- **Conditional probability:** P(A | B) = P(A ∩ B)/P(B).
- **Independent events:** P(A ∩ B) = P(A)P(B).
- **Random variable:** a function that assigns a number to each outcome; discrete (pmf) or continuous (pdf).
- **Expectation and variance:** E[X] = Σ x p(x) or ∫ x f(x) dx; Var(X) = E[X²] − (E[X])².
- **Central limit theorem (CLT):** the mean of n independent, identically distributed variables with finite variance is approximately normal for large n, with mean μ and variance σ²/n.

💡 Core concepts

Important distributions

Distribution	P(X = x) or f(x)	Mean	Variance
Bernoulli(p)	p^x(1 − p)^(1 − x)	p	p(1 − p)
Binomial(n, p)	C(n, x) p^x (1 − p)^(n − x)	np	np(1 − p)
Poisson(λ)	e^(−λ) λ^x/x!	λ	λ
Geometric(p)	(1 − p)^(x − 1) p	1/p	(1 − p)/p²

Distribution	P(X = x) or f(x)	Mean	Variance
Uniform(a, b)	1/(b − a)	(a + b)/2	(b − a)²/12
Exponential(λ)	λe^(−λx)	1/λ	1/λ²
Normal(μ, σ²)	(1/σ√(2π)) e^(−(x − μ)²/2σ²)	μ	σ²

Bayes theorem: $P(A_i \mid B) = P(B \mid A_i)P(A_i)/\Sigma_j P(B \mid A_j)P(A_j)$.

Inequalities

- Markov: for $X \geq 0$, $P(X \geq a) \leq E[X]/a$.
- Chebyshev: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$.

Correlation and regression

- $r = \Sigma(x - \bar{x})(y - \bar{y})/\sqrt{[\Sigma(x - \bar{x})^2 \Sigma(y - \bar{y})^2]}$.
- Regression of y on x: $y - \bar{y} = b_{yx}(x - \bar{x})$, $b_{yx} = r\,\sigma y/\sigma x$. Regression of x on y uses $b_{xy} = r\,\sigma x/\sigma y$. Also $r^2 = b_{yx} \cdot b_{xy}$.
- Spearman's rank correlation: $\rho = 1 - 6\Sigma d^2/[n(n^2 - 1)]$.
- Inference for correlation: $t = r\sqrt{(n - 2)}/\sqrt{(1 - r^2)}$ with $n - 2$ degrees of freedom.
- Multiple regression: $y = b0 + b1 \times 1 + \ldots + bkxk$; least squares estimate $b = (X^TX)^{-1}X^Ty$.

🧮 Solved examples

Example 1 (central tendency and spread). Data 2, 4, 4, 4, 5, 5, 7, 9.

- Mean = 40/8 = **5**; median = (4 + 5)/2 = **4.5**; mode = **4**.
- Population variance = $\Sigma(x - 5)^2/8 = 32/8 = 4$, so σ = **2**. Sample SD = $\sqrt{(32/7)} \approx 2.14$.
- CV = 2/5 × 100 = 40%. Mean > median > mode suggests slight positive skew.

Example 2 (correlation and regression). x = 1, 2, 3, 4, 5; y = 2, 4, 5, 4, 5.

- $\Sigma(x - \bar{x})(y - \bar{y}) = 6$, $\Sigma(x - \bar{x})^2 = 10$, $\Sigma(y - \bar{y})^2 = 6$.
- $r = 6/\sqrt{60} \approx$ **0.775** (fairly strong positive).
- Regression: b = 0.6, a = 2.2, so **y = 2.2 + 0.6x**.

Example 3 (distributions).

- Binomial n = 10, p = 0.5: $P(X = 3) = 120/1024 \approx$ **0.117**.
- Poisson λ = 2: $P(X = 0) = e^{-2} \approx$ **0.135**.
- Normal μ = 60, σ = 10: $P(X > 80) = P(Z > 2) \approx$ **0.0228**.

Example 4 (Chebyshev). With μ = 50 and σ = 5, at least $1 - 1/2^2 = 75\%$ of values lie between 40 and 60, whatever the distribution.

Example 5 (Bayes). Machines A, B, C make 50%, 30%, 20% of items with defect rates 1%, 2%, 3%. P(defective) = 0.005 + 0.006 + 0.006 = 0.017. $P(A \mid \text{defective}) = 0.005/0.017 \approx$ **0.294**.

Formulas to remember

- $\sigma^2 = \Sigma(x - \bar{x})^2/n$ (population), $s^2 = \Sigma(x - \bar{x})^2/(n - 1)$ (sample).

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- Sk = (mean − mode)/σ; β2 = μ4/μ2².
- r, b_yx, Spearman ρ as above.
- Var(X̄) = σ²/n (used in CLT).

⭐ Important questions

1. Explain measures of central tendency and dispersion; compute them for given data.
2. Explain skewness and kurtosis with formulas and diagrams.
3. Explain correlation (simple, multiple, rank) and regression; fit a regression line and test r.
4. State Bayes theorem and solve a problem; explain random variables, expectation and variance.
5. Explain binomial, Poisson and normal distributions; state Markov and Chebyshev inequalities and the CLT.

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UNIT 2 Inferential Statistics and Linear Methods

✍️ 2-mark definitions

- **Population and sample:** the population is the whole group of interest; a sample is a subset used to draw conclusions.
- **Parameter and statistic:** a parameter describes a population (μ, σ); a statistic is computed from a sample ($\bar{x}$, s).
- **Sampling distribution:** the distribution of a statistic over all possible samples; the standard error of $\bar{x}$ is $\sigma/\sqrt{n}$.
- **Point estimate vs interval estimate:** a single value ($\bar{x}$) vs a range that likely contains the parameter (confidence interval).
- **Confidence interval (CI):** an interval computed so that, in repeated sampling, a stated percentage (e.g. 95%) of such intervals contain the parameter.
- **Null hypothesis (H0) and alternative (H1):** H0 is the default claim (no effect); H1 is what we test for.
- **Type I and Type II errors:** rejecting a true H0 (probability α, significance level) and accepting a false H0 (probability β). Power = 1 − β.
- **p-value:** the probability, assuming H0, of a result at least as extreme as the one observed; reject H0 if $p < \alpha$.
- **Degrees of freedom:** the number of independent values free to vary when computing a statistic.
- **Goodness-of-fit test:** checks whether observed frequencies match a hypothesized distribution (chi-square).
- **Test of independence:** checks whether two categorical variables are related, using a contingency table (chi-square).
- **ANOVA:** analysis of variance; tests whether three or more group means are equal by comparing between-group and within-group variance.
- **Permutation (randomization) test:** a non-parametric test that builds the null distribution by reshuffling group labels.
- **Principal component analysis (PCA):** a linear transformation to new uncorrelated variables (principal components) ordered by variance, used for dimension reduction.
- **Singular value decomposition (SVD):** factorization $A = U\Sigma V^T$ with orthogonal U, V and non-negative singular values in Σ.

💡 Core concepts

Choosing a test

Situation	Test	Statistic
Mean, σ known or n large	z-test	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
Mean, σ unknown, small n	one-sample t-test	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$, df $n - 1$

Situation	Test	Statistic
Two independent means	two-sample t-test	$t = (\bar{x}1 - \bar{x}2)/(sp\sqrt{(1/n1 + 1/n2)})$
Before and after on same subjects	paired t-test	$t = \bar{d}/(s_d/\sqrt{n})$
Three or more means	one-way ANOVA	$F = MSB/MSW$
Observed vs expected counts	chi-square goodness of fit	$\chi^2 = \Sigma(O - E)^2/E$, df $k - 1$
Two categorical variables	chi-square independence	df $(r - 1)(c - 1)$

Steps of hypothesis testing: state H0 and H1 → choose α → compute the test statistic → find the critical value or p-value → decide (reject or fail to reject H0) → state the conclusion in words.

Confidence intervals: mean with σ known: $\bar{x} \pm z(\alpha/2)\ \sigma/\sqrt{n}$ ($z = 1.96$ for 95%, 2.576 for 99%); σ unknown: $\bar{x} \pm t(\alpha/2, n - 1)\ s/\sqrt{n}$; proportion: $\hat{p} \pm z\sqrt{(\hat{p}(1 - \hat{p})/n)}$.

Linear regression computation

- Normal equations $X^TXb = X^Ty$ can be ill-conditioned; numerically stable methods are preferred.
- **QR decomposition (Householder orthogonalization):** $X = QR$ with Q orthogonal and R upper triangular, built with Householder reflections $H = I - 2vv^T/(v^Tv)$. Then solve $Rb = Q^Ty$ by back substitution.
- **SVD:** $X = U\Sigma V^T$ gives $b = V\Sigma^+U^Ty$; it also handles rank-deficient X and shows multicollinearity through small singular values.
- **PCA:** centre the data, compute the covariance matrix (or SVD of the data matrix), take eigenvectors with the largest eigenvalues; the share of variance explained by component k is $\lambda k/\Sigma\lambda$.

🧮 Solved examples

Example 1 (95% CI). $n = 100$, $\bar{x} = 50$, $\sigma = 10$: SE = 1, CI = 50 ± 1.96 = **(48.04, 51.96)**.

Example 2 (one-sample t-test). $n = 16$, $\bar{x} = 52$, $s = 8$, H0: $\mu = 50$, $\alpha = 0.05$ (two-tailed). $t = 2/(8/4) =$ **1.0**; critical $t(0.025, 15) = 2.131$. Since $1.0 < 2.131$, **do not reject H0**.

Example 3 (chi-square goodness of fit). A die is rolled 60 times: observed 8, 9, 12, 11, 6, 14; expected 10 each. $\chi^2 = (4 + 1 + 4 + 1 + 16 + 16)/10 =$ **4.2**; df = 5; critical value 11.07 at 5%. $4.2 < 11.07$, so the die is **consistent with fair**.

Example 4 (one-way ANOVA). Groups A: 2, 3, 4 (mean 3); B: 4, 5, 6 (mean 5); C: 6, 7, 8 (mean 7). Grand mean 5.

- $SSB = 3[(3 - 5)^2 + 0 + (7 - 5)^2] = 24$, df 2, MSB = 12.
- $SSW = 2 + 2 + 2 = 6$, df 6, MSW = 1.
- $F = 12$; critical $F(2, 6)$ at 5% = 5.14. **Reject H0**: the group means differ.

Example 5 (PCA variance). Eigenvalues 4.0, 0.8, 0.2: the first component explains 4/5 = **80%** of the variance, the first two 96%.

Formulas to remember

- $SE = \sigma/\sqrt{n}$; $CI = \bar{x} \pm z\ SE$.

- z, t, F, χ^2 formulas in the table.
- Pooled variance $sp^2 = [(n1 - 1)s1^2 + (n2 - 1)s2^2]/(n1 + n2 - 2)$.
- Least squares $b = (X^TX)^{-1}X^Ty = R^{-1}Q^Ty$.

⭐ Important questions

1. Explain sampling distributions, point and interval estimation; build confidence intervals.
2. Explain hypothesis testing, type I and II errors and p-values; perform z and t tests (one-sample, independent, paired).
3. Explain chi-square tests for goodness of fit and independence, with examples.
4. Explain one-way ANOVA with an example; explain permutation and randomization tests.
5. Explain multiple regression and its computation by Householder QR and SVD; explain PCA for dimension reduction.

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UNIT 3 Random Numbers, Monte Carlo and MCMC

✍️ 2-mark definitions

- **Pseudo-random numbers:** numbers generated by a deterministic algorithm that look random; the sequence repeats after a period and depends on a seed.
- **Linear congruential generator (LCG):** $x_{n+1} = (a\,x_n + c) \bmod m$, $u_n = x_n/m$.
- **Inverse transform method:** if $U \sim \text{Uniform}(0, 1)$, then $X = F^{-1}(U)$ has CDF F.
- **Acceptance-rejection method:** generate Y from an easy density g, and accept it with probability $f(Y)/(c\,g(Y))$, where $f(x) \le c\,g(x)$ for all x.
- **Monte Carlo method:** estimating quantities (integrals, probabilities) by averaging over random samples.
- **Monte Carlo integration:** $\int_a^b h(x)\,dx \approx (b-a) \times (1/n) \sum h(U_i)$, U_i uniform on $[a, b]$.
- **Variance reduction:** techniques that make Monte Carlo estimates more precise for the same n: antithetic variables, control variates, importance sampling, stratified sampling.
- **Markov chain:** a sequence of random states where the next state depends only on the current one (Markov property).
- **Stationary distribution:** a distribution π with $\pi = \pi P$; a well-behaved chain converges to it.
- **MCMC (Markov chain Monte Carlo):** sampling from a complex distribution by building a Markov chain whose stationary distribution is the target.
- **Metropolis-Hastings:** an MCMC algorithm that proposes a move and accepts it with a probability based on the target density ratio.
- **Gibbs sampling:** an MCMC algorithm that updates one variable at a time from its full conditional distribution.
- **Burn-in:** the initial samples of a chain discarded before it reaches the stationary distribution.

💡 Core concepts

Variance reduction techniques

Technique	Idea
Antithetic variables	use U and $1 - U$ together; negative correlation lowers variance
Control variates	use a correlated variable C with known mean: $\hat{\theta} = \text{mean}(h) - c(\text{mean}(C) - E[C])$
Importance sampling	sample from g concentrated where $h \cdot f$ is large; weight by f/g
Stratified sampling	split the domain into strata and sample each separately

Metropolis-Hastings algorithm

1. Start at x_0.
2. Propose y from $q(y \mid x_t)$.
3. Compute $\alpha = \min(1, [\pi(y)\,q(x_t \mid y)]/[\pi(x_t)\,q(y \mid x_t)])$; for a symmetric q, $\alpha = \min(1, \pi(y)/\pi(x_t))$.

4. Draw U; if U ≤ α accept xt+1 = y, else xt+1 = xt.
5. Repeat; discard burn-in; use the rest as samples.

Gibbs sampler (two variables): repeat: draw x from f(x | y), then y from f(y | x). It is a special case of Metropolis-Hastings with acceptance probability 1.

Convergence diagnostics: trace plots, autocorrelation plots, running means, multiple chains from different starts (Gelman-Rubin statistic close to 1).

Monte Carlo hypothesis testing: simulate the test statistic many times under H0 and estimate the p-value as the proportion of simulated values at least as extreme as the observed one.

🧮 Solved examples

Example 1 (LCG). a = 5, c = 3, m = 16, x0 = 7: x1 = 38 mod 16 = 6; x2 = 33 mod 16 = 1; x3 = 8; x4 = 43 mod 16 = 11. Uniforms: 0.375, 0.0625, 0.5, 0.6875.

Example 2 (inverse transform for exponential). F(x) = 1 − e^(−λx), so X = −ln(1 − U)/λ (or −ln(U)/λ). With λ = 2 and U = 0.5: X = ln 2/2 ≈ **0.347**.

Example 3 (Monte Carlo integration). Estimate $\int_0^1 x^2\,dx = 1/3$. With U = 0.1, 0.4, 0.7, 0.9: mean of U^2 = (0.01 + 0.16 + 0.49 + 0.81)/4 = 0.3675 (close to 0.333; more samples give a better estimate, error ∝ 1/√n).

Example 4 (estimating π). Generate n points in the unit square; count those with $x^2 + y^2 \le 1$. π ≈ 4 × (count/n). If 785 of 1000 points fall inside, π ≈ 3.14.

Example 5 (Metropolis step). Target π(x) ∝ e^(−x²/2). Current x = 0, proposal y = 1: α = min(1, e^(−0.5)/1) ≈ 0.607. If U = 0.4 ≤ 0.607, accept y = 1.

⌨️ Code to remember (R)

```
set.seed(42)
# Inverse transform: exponential(rate = 2)
u <- runif(10000); x <- -log(u) / 2; mean(x)            # about 0.5

# Acceptance-rejection: Beta(2, 2) from Uniform, f(x) = 6x(1 - x) <= 1.5
ar <- function(n) { out <- c()
  while (length(out) < n) { y <- runif(1); if (runif(1) <= 6*y*(1-y)/1.5) out <- c(out, y) }
  out }

# Monte Carlo integral of exp(-x^2) on [0, 1] with antithetic variables
u <- runif(5000); est <- mean((exp(-u^2) + exp(-(1-u)^2)) / 2)

# Metropolis-Hastings for standard normal
mh <- function(n, sd = 1) { x <- numeric(n)
  for (t in 2:n) { y <- rnorm(1, x[t-1], sd)
    if (runif(1) < dnorm(y) / dnorm(x[t-1])) x[t] <- y else x[t] <- x[t-1] }
  x }
s <- mh(10000)[-(1:1000)]   # drop burn-in
```

⭐ Important questions

1. Explain pseudo-random number generation (LCG) and its properties.
2. Explain the inverse transform and acceptance-rejection methods with examples.
3. Explain Monte Carlo integration and Monte Carlo hypothesis testing.
4. Explain variance reduction: antithetic variables, control variates, importance and stratified sampling.
5. Explain Markov chains, the Metropolis-Hastings algorithm, Gibbs sampling and convergence checks.

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UNIT 4 Resampling, Density Estimation and Numerical Methods

✍️ 2-mark definitions

- **Resampling:** repeatedly drawing samples from the observed data to assess the variability of a statistic.
- **Bootstrap:** drawing B samples of size n **with replacement** from the data and recomputing the statistic each time; gives standard errors and confidence intervals.
- **Percentile bootstrap CI:** the interval between the $\alpha/2$ and $1 - \alpha/2$ percentiles of the bootstrap statistics.
- **Jackknife:** recomputing the statistic n times, each time leaving out one observation; estimates bias and standard error.
- **Cross-validation:** splitting data into training and validation parts to estimate prediction error; k-fold uses k parts in turn; leave-one-out uses k = n.
- **Permutation test:** reshuffling labels to obtain the null distribution of a statistic.
- **Density estimation:** estimating the probability density function from data without assuming a parametric form.
- **Histogram estimator:** counts in bins divided by n × bin width.
- **Kernel density estimator (KDE):** $\hat{f}(x) = (1/nh) \Sigma K((x - x_i)/h)$, a smooth sum of kernels centred at each point; h is the bandwidth.
- **Root finding:** finding x with $f(x) = 0$ (bisection, Newton-Raphson, secant).
- **Numerical integration:** approximating $\int f(x)\,dx$ (trapezoidal, Simpson's rule, Gaussian quadrature).
- **EM algorithm:** an iterative method for maximum likelihood with missing or latent data, alternating an E-step and an M-step.
- **Simplex method:** an algorithm for linear programming that moves along the vertices of the feasible region to the optimum. (Nelder-Mead "simplex" is a different, derivative-free method for unconstrained optimization.)

💡 Core concepts

Bootstrap vs jackknife

Point	Bootstrap	Jackknife
Samples	B (e.g. 1000) random resamples with replacement	n leave-one-out samples
Randomness	yes	no (deterministic)
Gives	SE, bias, CIs, distribution	mainly bias and SE
Works for median	yes	poorly (non-smooth statistics)

Jackknife formulas: $\hat{\theta}(\cdot)$ = mean of the leave-one-out estimates; bias = $(n - 1)(\hat{\theta}(\cdot) - \hat{\theta})$; SE = $\sqrt{[((n - 1)/n) \Sigma(\hat{\theta}(i) - \hat{\theta}(\cdot))^2]}$.

Kernel choice and bandwidth: Gaussian, Epanechnikov and uniform kernels; the bandwidth matters far more than the kernel. Silverman's rule of thumb: h = 1.06 σ n^(−1/5). Small h → wiggly (high variance); large h → over-smoothed (high bias).

Root finding methods

Method	Update	Convergence
Bisection	halve the interval where the sign changes	linear, always converges
Newton-Raphson	$x_1 = x_0 - f(x_0)/f'(x_0)$	quadratic, needs derivative and good start
Secant	uses two previous points instead of f′	superlinear

Numerical integration

- Trapezoidal: ∫ ≈ (h/2)[f0 + 2(f1 + … + fn−1) + fn].
- Simpson's 1/3 (n even): ∫ ≈ (h/3)[f0 + 4(f1 + f3 + …) + 2(f2 + f4 + …) + fn].

Optimization: maximization or minimization by Newton's method (uses the Hessian), gradient descent, Nelder-Mead; constrained problems use Lagrange multipliers, penalty methods or linear programming (simplex). In R: optimize() (1-D), optim(), nlm(), and constrOptim().

EM algorithm (mixture of two normals): E-step: compute each point's responsibility (probability of belonging to each component) using current parameters. M-step: update the means, variances and mixing proportions as responsibility-weighted averages. Repeat until the log-likelihood stops increasing; the likelihood never decreases.

🧮 Solved examples

Example 1 (jackknife mean). Data 2, 4, 6: $\hat{\theta} = 4$. Leave-one-out means: 5, 4, 3; $\hat{\theta}(\cdot) = 4$; bias = 0 (the mean is unbiased). SE = $\sqrt{[(2/3)(1 + 0 + 1)]} = \sqrt{(4/3)} \approx$ **1.155**, which equals s/√n.

Example 2 (Newton-Raphson for √2). $f(x) = x^2 - 2$, x0 = 1: x1 = 1 − (−1)/2 = 1.5; x2 = 1.5 − 0.25/3 ≈ 1.41667; x3 ≈ **1.41422**.

Example 3 (numerical integration of $\int_0^1 x^2\,dx = 1/3$, h = 0.5).

- Trapezoidal: (0.5/2)(0 + 2 × 0.25 + 1) = **0.375**.
- Simpson's: (0.5/3)(0 + 4 × 0.25 + 1) = **0.3333** (exact for polynomials up to degree 3).

Example 4 (k-fold CV). With 100 observations and k = 5, each fold has 20 points; the model is trained 5 times on 80 and tested on 20, and the 5 errors are averaged.

Example 5 (bisection). $f(x) = x^3 - x - 2$ on [1, 2]: f(1) = −2, f(2) = 4 → midpoint 1.5: f = −0.125 → root in [1.5, 2] → 1.75: f = 1.609 → [1.5, 1.75] → 1.625 … converging to 1.5214.

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⌨️ Code to remember (R)

```r
x <- c(12, 15, 9, 20, 17, 11, 14)
# Bootstrap SE and percentile CI of the median
B <- 2000; meds <- replicate(B, median(sample(x, replace = TRUE)))
sd(meds); quantile(meds, c(0.025, 0.975))

# Jackknife SE of the mean
n <- length(x); th <- sapply(1:n, function(i) mean(x[-i]))
sqrt((n - 1) / n * sum((th - mean(th))^2))

# Kernel density estimate and root finding
d <- density(x, bw = "nrd0"); plot(d)
uniroot(function(z) z^3 - z - 2, c(1, 2))$root      # 1.5214
integrate(function(z) z^2, 0, 1)                    # 0.3333
optim(c(1, 1), function(p) (p[1] - 3)^2 + (p[2] + 1)^2)$par   # about (3, -1)
```

⭐ Important questions

1. Explain the bootstrap and percentile confidence intervals; explain the jackknife with formulas.
2. Explain cross-validation and permutation tests.
3. Explain histogram and kernel density estimation; discuss bandwidth selection and multivariate density estimation.
4. Explain root finding (bisection, Newton-Raphson) and numerical integration (trapezoidal, Simpson) with examples.
5. Explain constrained and unconstrained optimization, the EM algorithm and the simplex method.

UNIT 5 Statistical Computing with R

✍️ 2-mark definitions

- **R:** a free, open-source language and environment for statistical computing and graphics, created by Ross Ihaka and Robert Gentleman (University of Auckland, 1993) as an implementation of the S language.
- **CRAN:** the Comprehensive R Archive Network, which hosts R and thousands of packages.
- **Package:** a bundle of R functions, data and documentation; installed with install.packages("name") and loaded with library(name).
- **Workspace:** the current R environment holding user objects; ls() lists them, rm() removes them, save.image() saves them.
- **Vector:** the basic R object; an ordered collection of values of one type, created with c(), seq() or rep().
- **Matrix and array:** 2-D and n-D collections of one type, created with matrix() and array().
- **List:** an ordered collection of objects of possibly different types.
- **Data frame:** a table where columns may have different types but equal length; the standard structure for datasets.
- **Factor:** a categorical variable with a fixed set of levels.
- **Function:** a reusable block of code: f ← function(args) { body }.
- **Vectorization:** applying an operation to a whole vector at once, which is faster than loops in R.
- **apply family:** apply (rows/columns of a matrix), lapply (list, returns list), sapply (simplifies), tapply (by groups), mapply (multiple inputs).

💡 Core concepts

Data types: numeric (double), integer (5L), character, logical (TRUE/FALSE), complex, and raw. Check with class(), typeof(), is.numeric(); convert with as.numeric(), as.character(), as.factor().

Operators: arithmetic (+ − * / ^ %% %/%), relational (< > == != <= >=), logical (& | ! && ||), assignment (← = →), and %in% for membership; matrix multiply %*%.

Flow control: if / else, ifelse(test, yes, no) (vectorized), for, while, repeat with break, next, and switch().

Dates and times: as.Date("2024-01-15"), format(d, "%d/%m/%Y"), Sys.Date(), difftime(), seq(d, by = "month", length.out = 6); POSIXct for date-times.

Import and export: read.csv(), read.table(), readLines(), readRDS(); write.csv(), write.table(), saveRDS(), sink() for output to a file.

Graphics: plot(), hist(), boxplot(), barplot(), pie(), lines(), points(), abline(), legend(), text(); par(mfrow = c(2, 2)) for multiple plots; export with png("f.png"), pdf("f.pdf") and dev.off(); interactive annotation with locator() and identify().

Descriptive statistics and aggregation: summary(), mean(), median(), var(), sd(), quantile(), range(), table(), cor(); aggregate(y ~ g, data, FUN = mean), tapply(), by(), and multivariate views with pairs() and cor() matrices.

Code optimization: vectorize, pre-allocate vectors (numeric(n)) instead of growing them, use apply functions, profile with system.time() and Rprof(), and move repeated code into functions (code factorization).

Statistical functions: t.test(), chisq.test(), aov(), lm(), glm(), cor.test(), wilcox.test(), prcomp() (PCA), and distribution functions d/p/q/r + name (dnorm, pnorm, qnorm, rnorm).

🧮 Solved examples

Example 1 (vectors and indexing).

```
v <- c(4, 8, 15, 16, 23, 42)
v[2]            # 8
v[-1]           # 8 15 16 23 42
v[v > 15]       # 16 23 42
sum(v); mean(v) # 108  18
seq(1, 10, by = 3)   # 1 4 7 10
rep(c("a", "b"), times = 2)   # "a" "b" "a" "b"
```

Example 2 (matrix).

```
m <- matrix(1:6, nrow = 2)   # filled by column: [1 3 5; 2 4 6]
dim(m); t(m)                  # 2 3, transpose
m %*% t(m)                    # [35 44; 44 56]
apply(m, 1, sum)              # row sums 9 12
```

Example 3 (data frame and factor).

```
df <- data.frame(name = c("Asha", "Ravi", "Neha"), marks = c(82, 67, 91),
                 branch = factor(c("CSE", "IT", "CSE")))
df[df$marks > 70, ]              # Asha and Neha
levels(df$branch)                # "CSE" "IT"
aggregate(marks ~ branch, df, mean)   # CSE 86.5, IT 67
```

Example 4 (user-defined function and loop).

```
grade <- function(m) if (m >= 90) "A" else if (m >= 75) "B" else "C"
for (m in c(95, 80, 60)) print(grade(m))   # "A" "B" "C"
sapply(c(95, 80, 60), grade)              # vectorized form
```

⌨️ Code to remember (R)

```r
# Read, summarize, test, model and plot
data <- read.csv("students.csv")          # columns: hours, score, group
str(data); summary(data)
t.test(score ~ group, data = data)         # two-sample t-test
chisq.test(table(data$group, data$score > 60))
fit <- lm(score ~ hours, data = data); summary(fit); coef(fit)
png("scatter.png"); plot(data$hours, data$score, main = "Hours vs Score")
abline(fit, col = "red"); dev.off()
write.csv(aggregate(score ~ group, data, mean), "group_means.csv", row.names = FALSE)
```

⭐ Important questions

1. Explain the history and features of R; explain packages, the workspace and basic operators.
2. Explain R data objects: vectors, matrices, arrays, lists, data frames and factors, with examples.
3. Explain functions, flow control and loops in R; write a user-defined function.
4. Explain importing and exporting data, graphics and exporting graphs in R.
5. Explain descriptive statistics, aggregation and statistical tests in R; explain code factorization and optimization.

Last-hour checklist

Read only this in the last hour before the paper.

- ☐ Mean, median, mode; σ vs s; CV; skewness and kurtosis formulas.
- ☐ r = 0.775 example; regression lines; Spearman; t for r.
- ☐ Bayes (machine defect example); distribution table (mean and variance).
- ☐ Markov and Chebyshev; CLT statement.
- ☐ CI formula; hypothesis testing steps; type I and II errors; p-value.
- ☐ t-test (t = 1.0 example), chi-square (4.2), ANOVA (F = 12).
- ☐ QR (Householder), SVD, PCA variance explained.
- ☐ LCG; inverse transform; acceptance-rejection; Monte Carlo integration and π.
- ☐ Variance reduction table; Metropolis-Hastings acceptance; Gibbs; burn-in.
- ☐ Bootstrap vs jackknife; cross-validation; KDE and bandwidth.
- ☐ Newton-Raphson (√2), bisection, trapezoidal 0.375 vs Simpson 0.3333; EM; simplex.
- ☐ R objects, apply family, import/export, plots, t.test, chisq.test, lm.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Fundamentals of Mathematical Statistics	S. C. Gupta, V. K. Kapoor	Sultan Chand & Sons	WorldCat
2	Introduction to Probability and Statistics for Engineers and Scientists	Sheldon M. Ross	Academic Press	WorldCat
3	Modern Mathematical Statistics	E. J. Dudewicz, S. N. Mishra	Wiley	WorldCat
4	Statistics Using R	S. G. Purohit, S. D. Gore, S. R. Deshmukh	Narosa	WorldCat
5	Statistical Computing with R	Maria L. Rizzo	Chapman & Hall/CRC	WorldCat
6	The Art of R Programming	Norman Matloff	No Starch Press	WorldCat
7	Introductory Statistics with R	Peter Dalgaard	Springer	WorldCat
8	Probability and Statistics with R	M. D. Ugarte, A. F. Militino, A. T. Arnholt	CRC Press	WorldCat
9	Statistical Computing: Existing Methods and Recent Developments	D. Kundu, A. Basu	Narosa	WorldCat
10	Handbook of Computational Statistics	J. E. Gentle, W. K. Härdle, Y. Mori	Springer	WorldCat
11	Computational Statistics	G. H. Givens, J. A. Hoeting	Wiley	WorldCat
12	The R Book	Michael J. Crawley	John Wiley & Sons	WorldCat
13	Learning R	Richard Cotton	O'Reilly	WorldCat
14	A Handbook of Statistical Analyses Using R (2nd ed.)	Brian S. Everitt, Torsten Hothorn	Chapman & Hall/CRC	WorldCat
15	Learning Statistics Using R	Randall E. Schumacker	Sage	WorldCat
16	R for Everyone	Jared P. Lander	Addison Wesley	WorldCat
17	Numerical Methods of Statistics	John F. Monahan	Cambridge University Press	WorldCat
18	Introducing Monte Carlo Methods with R	Christian Robert, George Casella	Springer	WorldCat

The R Project website (r-project.org) and CRAN manuals are free, legal companions for Unit 5.

Syllabus source: AKTU B.Tech 3rd year CS/IT syllabus, 2024-25 (copy hosted by an affiliated college, PDF)